

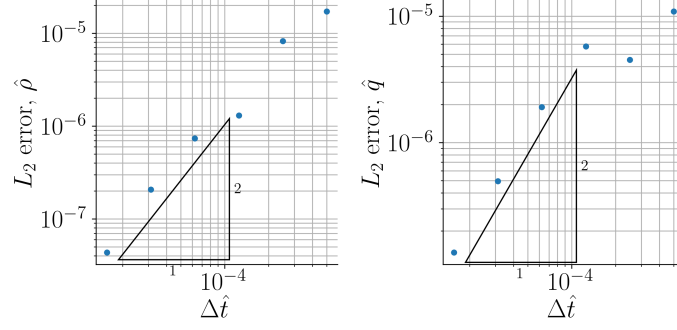


FIG. 9. Second-order convergence of the numerical solution of the fields $\hat{\rho}$ and $\hat{q}$.

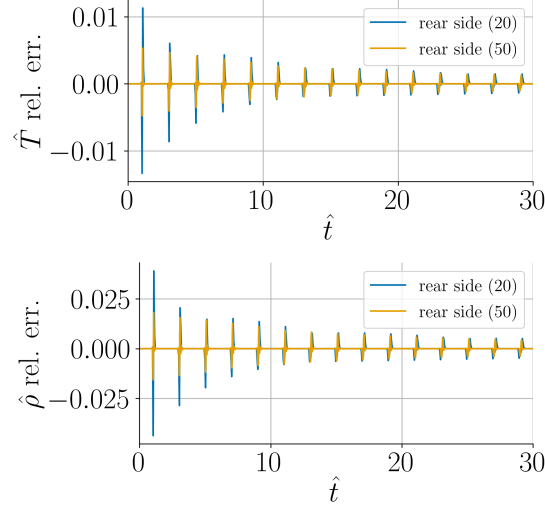


FIG. 10. Grid dependence of time evolution of temperature and density at the rear side of the sample.

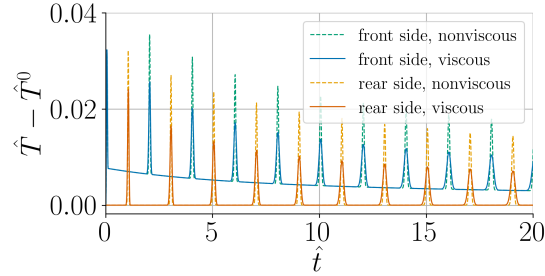


FIG. 11. The effect of viscous momentum transport on the temperature field.

B. The thermoacoustic approximation of the piston effect

This section is devoted to analyze the piston effect in supercritical fluids via the developed scheme. In this case the duration of excitation is much longer than the acoustic time scale. Therefore, wave propagation occurs only at the beginning of the process until the initial pressure disturbance reaches the other end of the pipe, correspondingly, no reflection takes place. In order to prove the intensive effect of thermal expansion near the critical point, we present the same phenomenon with same dimensional excitation parameters for an ideal gas state, too.

Based on [19], we have defined a cross-section specific heat of $q = 30 \frac{\text{J}}{\text{m}^2}$, to which in the above defined supercritical state the dimensionless value $\hat{q} = 0.00035$ corresponds, the non-dimensional pulse duration $\hat{t}_p = 50$ and Péclet number $\mathcal{Pe}_a = 10^7$ are chosen, which latter implies $\mathcal{Re}_a = 1722695.711$. Now, viscous momentum transport is also taken into